

Question 1.**Marks**

- (a) Find (i) $\int \sin(ax + \frac{\pi}{6}) dx$. 1
- (ii) $\int \sin^2(ax + \frac{\pi}{6}) dx$, where a is a constant. 2
- (b) Find $\int \frac{dx}{\sqrt{x^2 - 6x + 13}}$. 3
- (c) Evaluate $\int_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \sin^{-1}(x) dx$. 3
- (d) (i) Given that $\frac{2x^2 - 2x + 1}{(x+1)(2x^2 + 3)} = \frac{1}{x+1} + \frac{ax+b}{2x^2 + 3}$, 2
- find the real numbers a and b .
- (ii) Hence, or otherwise evaluate $\int_0^{\frac{\sqrt{3}}{\sqrt{2}}} \frac{2x^2 - 2x + 1}{(x+1)(2x^2 + 3)} dx$. 4

Question 2.**[START A NEW PAGE]**

- (a) Given $f(x) = x^2 - 2x$, sketch the graph of the following on separate axes
- (i) $y = f(x)$. 1
- (ii) $y = f(x) + |f(x)|$. 2
- (iii) $y^2 = f(x)$. 2
- (iv) $y = [f(x)]^{-1}$. 2
- (v) $y = f(2^x)$. 2
- (b) (i) For the Hyperbola: $H: \frac{x^2}{9} - \frac{y^2}{16} = 1$, find its eccentricity e . 2
- (ii) Hence, neatly sketch the graph of H , clearly showing the vertices, foci, directrices and asymptotes. 4

Question 3.**[START A NEW PAGE]****Marks**

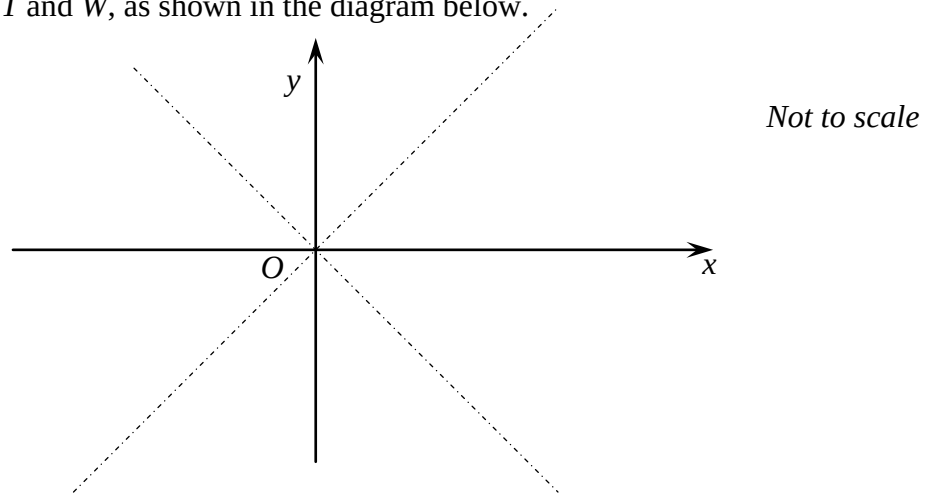
- (a) An ellipse E has the equation: $E: \frac{x^2}{100} + \frac{y^2}{75} = 1$.
- (i) Sketch this ellipse, clearly showing on your diagram the coordinates of the foci and the equation of each directrix. **4**
- (ii) Show that the equation of the normal to the ellipse E is $4x - 2y = 5$ at point $P(5, 7\frac{1}{2})$. **2**
- (iii) A circle is tangential to the ellipse E at P and at $Q(5, -7\frac{1}{2})$. **1**
Show that the centre of the circle is $(1\frac{1}{4}, 0)$.
- (iv) Hence find the equation of this circle. **2**
- (b) Using the substitution: $t = \tan \frac{x}{2}$, where $0 \leq x < \pi$, show that: **4**
- $$\int_0^{\frac{2\pi}{3}} \frac{dx}{5 + 3 \sin x + 4 \cos x} = \frac{\sqrt{3} - 1}{3}.$$
- (c) For $x > 0$, find: $\int \frac{dx}{x^4 + x}$. **2**

Question 4.

[START A NEW PAGE]

Marks

- (a) The point $S(3e, 0)$ is a focus of the hyperbola $H: x^2 - y^2 = 9$.
The tangent to the hyperbola, at the point $P(p, q)$, meets the asymptotes of H in T and W , as shown in the diagram below.



- (i) Show that the equation of the tangent TW is given by: $px - qy = 9$. **1**
- (ii) Show that the gradient of the line through SW is given by **1**
- $$m_{SW} = \frac{3}{e(p+q)-3}.$$
- (iii) By letting $\angle WST = \theta$, find the value of $\tan \theta$. **3**

- (b) Let $I_n = \int_0^{\frac{\pi}{4}} \tan^n x \, dx$, for $n = 0, 1, 2, \dots$

- (i) Show that $I_1 = \frac{1}{2} \ln 2$. **2**
- (ii) Show that, $I_n + I_{n-2} = \frac{1}{n-1}$, for $n = 2, 3, 4, \dots$ **3**
- (iii) Explain why $I_n < I_{n-2}$, for $n \geq 2$, and deduce that **3**

$$\frac{1}{2n+2} < I_n < \frac{1}{2n-2}.$$

- (iv) By using the recurrence relation in part (b) (ii), **2**
find I_5 , and deduce that: $\frac{2}{3} < \ln 2 < \frac{3}{4}$.

THE END



Question 1

i) $\int \sin(ax + \frac{\pi}{6}) dx = -\frac{1}{a} \cos(ax + \frac{\pi}{6}) + C$ ①

ii) $\int \sin^2(ax + \frac{\pi}{6}) dx$
 $= \frac{1}{2} \int [1 - \cos 2(ax + \frac{\pi}{6})] dx$ ①

$= \frac{x}{2} - \frac{1}{4a} \sin 2(ax + \frac{\pi}{6}) + C$ ①

iii) $\int \frac{1}{x^2 - 6x + 13} dx$

$= \int \frac{1}{(x-3)^2 + 4} dx$ ①

$= \frac{1}{2} \tan^{-1}(\frac{x-3}{2}) + C$ ①

$\int_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \sin^{-1} x dx = [x \sin^{-1} x]_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} - \int_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \frac{x}{\sqrt{1-x^2}} dx$

$= (\frac{\sqrt{3}}{2} \sin^{-1} \frac{\sqrt{3}}{2} - \frac{1}{2} \sin^{-1} \frac{1}{2}) + \int_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \frac{d(1-x)}{\sqrt{1-x^2}}$

$= \frac{\sqrt{3}\pi}{6} - \frac{\pi}{12} + \frac{1}{2} [(1-x^2)^{-\frac{1}{2}}]_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}}$

$= \frac{2\sqrt{3}\pi - \pi + 3 - 3\sqrt{3}}{12}$

i) $\frac{2x^2 - 2x + 1}{(x+1)(2x^2+3)} = \frac{1}{x+1} + \frac{ax+b}{2x^2+3}$

RHS. $= \frac{2x^2+3+(ax+b)(x+1)}{(x+1)(2x^2+3)}$ ①
 $= \frac{(2+a)x^2 + x(a+b) + b+3}{(x+1)(2x^2+3)}$

equating coefficients:

$2=2+a$
 $-2=a+b$
 $1=b+3$
 $\therefore a=0$
 and $b=-2$ ①

ii) $\int_0^{\frac{\sqrt{3}}{2}} [\frac{1}{x+1} + \frac{-2}{2x^2+3}] dx$ ①

$= [\ln(x+1) + \frac{-1}{\sqrt{3}} \tan^{-1}(\frac{\sqrt{3}}{3}x)]_0^{\frac{\sqrt{3}}{2}}$

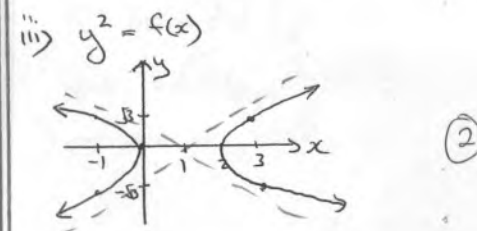
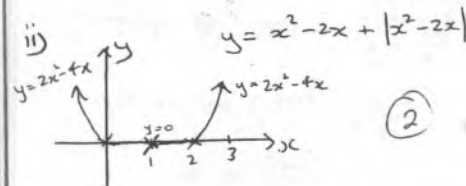
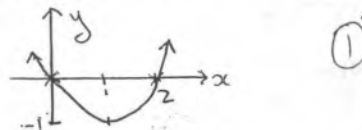
$= [\ln(x+1)]_0^{\frac{\sqrt{3}}{2}} + \frac{-\sqrt{3}}{3} [\tan^{-1} \frac{\sqrt{3}x}{\sqrt{3}}]_0^{\frac{\sqrt{3}}{2}}$ ①

$= \ln(\frac{\sqrt{3}}{2}+1) + \frac{-\sqrt{3}}{3} (\tan^{-1} 1 - \tan^{-1} 0)$ ①

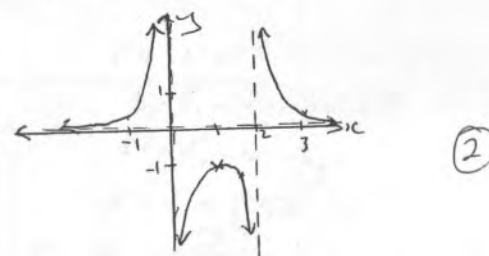
$= \ln(\frac{\sqrt{3}}{2}+1) - \frac{\sqrt{3}}{3} \frac{\pi}{4}$ ①

Question 2

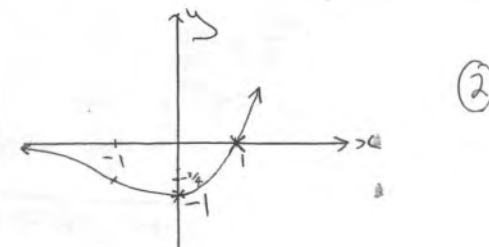
i) $y = x^2 - 2x$



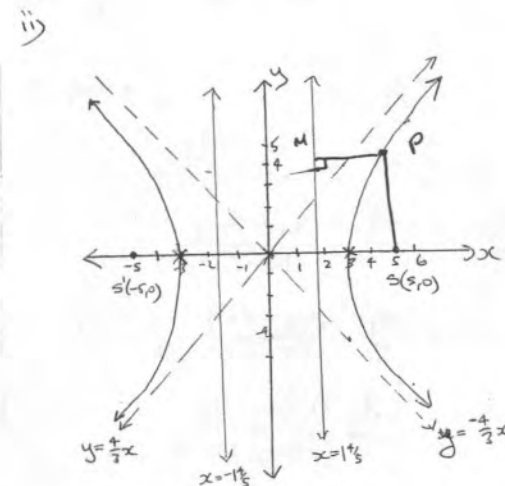
iv) $y = [f(x)]^{-1} = \frac{1}{x^2 - 2x}$



v) $y = f(2^x) = 2^{2x} - 2^{x+1}$



b) i) $b^2 = a^2(e^2 - 1)$ ①
 $16 = 9(e^2 - 1)$ ①
 $\frac{16}{9} + 1 = e^2$ ①
 $e = \frac{5}{3}$ ①



- ① for directrices
- ① for vertices
- ① for asymptotes
- ① for focal shape

Question 2

a) $\frac{x^2}{100} + \frac{y^2}{75} = 1$

$a=10 \quad b=\sqrt{75}=5\sqrt{3}$

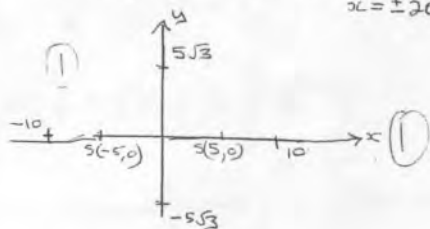
$b^2 = a^2(1-e^2)$

$\frac{75}{100} = 1-e^2$

$e^2 = \frac{1}{4}$

$e = \frac{1}{2}$

* foci are $(\pm 5, 0)$
* directrices are $x = \pm 20$



ii) Differentiate with respect to x :

$\frac{2x}{100} + \frac{2y}{75} \frac{dy}{dx} = 0$

$\frac{dy}{dx} = -\frac{x}{50} \cdot \frac{75}{2y} = -\frac{3x}{4y}$

① At $P(5, 7\frac{1}{2})$ $\frac{dy}{dx} = \frac{-3(5)}{4(7\frac{1}{2})} = -\frac{1}{2}$

∴ gradient of normal is 2.

eqn. of normal is:

$y - 7\frac{1}{2} = 2(x - 5)$

$y = 2x - 2\frac{1}{2}$

$2y = 4x - 5$

ie $x - 2y = 5$

iii) The intersection of the normals at P and Q will lie on the x-axis, the centre of the tangential circle is point of intersection of these 2 normals.

① sub. in $y=0$ (lies on x-axis)

iv) radius $= \sqrt{(5-1\frac{1}{4})^2 + (7\frac{1}{2}-0)^2}$
 $= \sqrt{3\frac{3}{4}^2 + 7\frac{1}{2}^2}$
 $= \sqrt{\frac{225}{16} + \frac{900}{16}}$
 $= \sqrt{\frac{1125}{4}}$

eqn. of circle is:

$(x - 1\frac{1}{4})^2 + y^2 = \frac{1125}{16}$

$(4x - 5)^2 + 16y^2 = 1125$

b) $\int_0^{2\pi} \frac{dx}{5 + 3\sin x + 4\cos x}$

$t = \tan \frac{x}{2}$ when $x=0, t=0$
 $x=2\pi, t=\sqrt{3}$

$\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$

∴ $dx = 2\cos^2 \frac{x}{2} dt$

∴ $\int_0^{\sqrt{3}} \frac{\frac{2}{1+t^2} dt}{5 + \frac{6t}{1+t^2} + \frac{4-t^2}{1+t^2}} = \int_0^{\sqrt{3}} \frac{2 dt}{5 + 5t^2 + 4t - t^2}$

$= 2 \int_0^{\sqrt{3}} \frac{dt}{t^2 + 4t + 9} = 2 \int_0^{\sqrt{3}} \frac{dt}{(t+3)^2 + 2}$

① $= -2 \left[\frac{1}{t+3} \right]_0^{\sqrt{3}}$

$= \frac{-2}{\sqrt{3}+3} + \frac{2}{3}$

$= \frac{-2\sqrt{3}+6}{3-9} + \frac{2}{3}$

$= \frac{2\sqrt{3}-2}{6}$

$= \frac{\sqrt{3}-1}{3}$ Q.E.P.

c) $\int \frac{dx}{x^2+x}$

$= \int \frac{1}{x^2+x} dx$

$= -\frac{1}{2} \int \frac{du}{u} = -\frac{1}{2} \ln(1+u) + C$

Question 4

a) $x^2 - y^2 = 9$

$2x - 2y \frac{dy}{dx} = 0$

∴ $\frac{dy}{dx} = \frac{x}{y}$

at $P(p, q)$ $\frac{dy}{dx} = \frac{p}{q}$

eqn. of tangent at P is:

$y - q = \frac{p}{q}(x - p)$

$qy - q^2 = px - p^2$

$p^2 - q^2 = px - qy$

but P lies on the Hyperbola

∴ $p^2 - q^2 = 9$

∴ tangent is $px - qy = 9$

Q.E.P.

ii) W lies on tangent at P and

the asymptote $y = -x$

∴ $px + qx = 9$

$x = \frac{9}{p+q}$

∴ W is $(\frac{9}{p+q}, \frac{-9}{p+q})$

Gradient of SW = $\frac{0 - \frac{-9}{p+q}}{3e - \frac{9}{p+q}}$

$= \frac{9}{3e(p+q) - 9}$

$= \frac{3}{e(p+q) - 3}$

iii) T lies on $y = x$ and on the tangent at P

∴ $px - qx = 9$

$x = \frac{9}{p-q}$

∴ T is $(\frac{9}{p-q}, \frac{9}{p-q})$

Grad. of ST = $\frac{3}{3-e(p-q)}$

If $\angle WST = \theta$ then

$\tan \theta = \frac{m_2 - m_1}{1 + m_1 m_2}$

$\tan \theta = \frac{\frac{3}{e(p+q)-3} - \frac{3}{3-e(p-q)}}{1 + \frac{9}{[e(p+q)-3][3-e(p-q)]}}$

$\tan \theta = \frac{3[3-e(p-q)] - 3[e(p+q)-3]}{[e(p+q)-3][3-e(p-q)] + 9}$
 $= \frac{9 - 3ep + 3eq - 3ep - 3eq + 9}{(ep+eq-3)(3-ep+eq) + 9}$
 $= \frac{18 - 6ep}{3ep - e^2p^2 + e^2pq + 3eq - e^2q^2 + 9}$

$= \frac{6(3-ep)}{6ep - e^2(p^2 - q^2)}$

① $= \frac{6(3-ep)}{6ep - 9e^2} [as p^2 - q^2 = 9]$

$= \frac{6(3-ep)}{6ep - 18} [as e^2 = 2]$

$= \frac{6(3-ep)}{6(ep-3)}$ $p \neq \frac{3}{e} = \frac{3}{\sqrt{2}}$

∴ $\tan \theta = -1$

b) I_1 is when $n=1$

$I_1 = \int_0^{\pi/4} \tan x dx$

$= \int_0^{\pi/4} \frac{\sin x}{\cos x} dx$

$= [-\ln(\cos x)]_0^{\pi/4}$

$= -\ln \cos \frac{\pi}{4} + \ln \cos 0$
 $= -\ln \frac{1}{\sqrt{2}} + \ln 1$

b) cont. (4/16)

ii) show $I_n + I_{n-2} = \frac{1}{n-1}$

$$\text{L.H.S.} = I_n + I_{n-2} = \int_0^{\pi/4} (\tan^n x) dx + \int_0^{\pi/4} \tan^{n-2} x dx$$

$$= \int_0^{\pi/4} (\tan^n x + \tan^{n-2} x) dx \quad (1)$$

$$= \int_0^{\pi/4} \tan^{n-2} x (\tan^2 x + 1) dx$$

$$= \int_0^{\pi/4} \tan^{n-2} x \cdot \sec^2 x dx \quad (1)$$

$$= \left[\frac{1}{n-1} \tan^{n-1} x \right]_0^{\pi/4}$$

$$= \frac{1}{n-1} [\tan^{n-1} \frac{\pi}{4} - \tan^{n-1} 0] \quad (1)$$

$$= \frac{1}{n-1} (1 - 0)$$

$$= \frac{1}{n-1}$$

$$\text{R.H.S.}$$

iii) let area represent $I_n \therefore A_n = I_n$

$$\text{now } 0 \leq x \leq \frac{\pi}{4}$$

$$\therefore 0 \leq \tan x \leq 1$$

$$\text{so } 0 \leq \tan^n x \leq \tan^2 x < \tan x \leq 1$$

$$\text{i.e. } 0 \leq \tan^{n+2} x < \tan^n x < \tan^{n-2} x < \dots < \tan^2 x < \tan x \leq 1 \quad \text{for } n \geq 2$$

$$\text{now for } \int_0^{\pi/4} \dots dx$$

becomes

$$0 \leq I_{n+2} < I_n < I_{n-2} < \dots < I_2 < I_1 \leq \frac{\pi}{4} \quad (1)$$

$$I_n < I_{n-2}$$

$$\text{now } I_{n+2} < I_n < I_{n-2}$$

$$I_{n+2} + I_n < 2I_n < I_n + I_{n-2}$$

$$\frac{1}{n+1} < 2I_n < \frac{1}{n-1}$$

$$\frac{1}{2(n+1)} < I_n < \frac{1}{2(n-1)} \quad \text{for } n \geq 2 \quad (1)$$

iv) when $n=5 \Rightarrow \frac{1}{10+2} < I_5 < \frac{1}{10-2}$

$$\therefore \frac{1}{12} < I_5 < \frac{1}{8}$$

$$\text{now } I_5 + I_3 = \frac{1}{4} \quad (\text{from ii})$$

$$I_3 + I_1 = \frac{1}{2}$$

$$I_3 + \frac{1}{2} \ln 2 = \frac{1}{2} \quad (\text{as } I_1 = \frac{1}{2})$$

$$I_3 = \frac{1}{2} - \frac{1}{2} \ln 2 \quad (1)$$

$$\therefore I_5 = \frac{1}{4} - \left(\frac{1}{2} - \frac{1}{2} \ln 2 \right)$$

$$= -\frac{1}{4} + \frac{1}{2} \ln 2$$

$$\text{so } \frac{1}{12} < -\frac{1}{4} + \frac{1}{2} \ln 2 < \frac{1}{8} \quad (1)$$

$$\frac{1}{3} < \frac{1}{2} \ln 2 < \frac{3}{8}$$

$$\therefore \frac{2}{3} < \ln 2 < \frac{3}{4}$$

